Course: Individual Software Development Process' 2026

Software Requirement Specification

By Guild Board

Chosen Irl Challenge: **Project D**: Course Support & Activity Dashboard

GUILD BOARD

Project D: COURSE SUPPORT & ACTIVITY DASHBOARD

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Table of Contents

1. Target Users

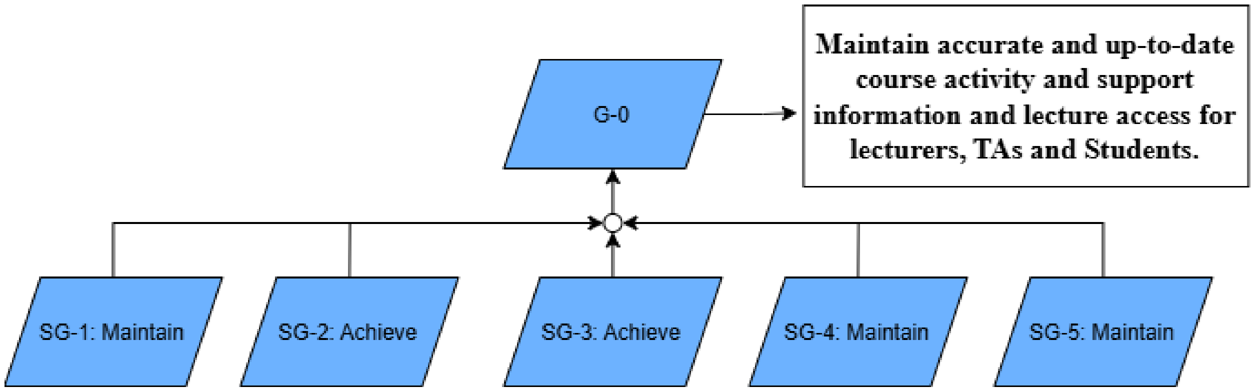
- **Lecturers:** Get an integrated view of course activities, milestones, and at-risk items across the semester; delegate or reassign tasks to TAs.
- **TAs (Teacher Assistance):** Log and update their assigned tasks (grading, lab prep, consultation slots), track small internal tickets, and see what's due when.
- **System administrator:** Maintain structured records of tasks, events and manage access to administrative funtions.
- **Students:** View the course activity calendar, deadlines, and up-to-date course materials/study plan published by the lecturer. (Read-only access, no ability to create or edit content.)

2. The System Goal

G-0: Maintain accurate and up-to-date course activity and support information and lecture access for lecturers, TAs and Students.

This Course Support & Activity Dashboard is used to help all target users achieve, avoid, and maintain their activities and schedules for continuing education and workflow. Course coordination at [department/course] currently relies on outdated tools: deadlines are sometimes not announced to users, and schedules are difficult to understand or act on without any shared plan within the tools. This causes issues to compound rapidly, as no record is kept of ongoing work or who is responsible for it. As a result, TAs and lecturers face communication breakdowns and lack a shared plan, leaving students unable to follow the current situation and these issues go unresolved because there is no shared history to refer back to.But for this iteration of project, the system scope will be limited to course activity coordination (calendar, materials, and task/ticket tracking) and support visibility (at-risk flagging, consultation logging, and an administrator-facing data overview). Student self-service booking, real-time messaging/notifications, grading, and integration with the university's LMS are explicitly out of scope for this iteration and will be addressed in later iterations.

3. Goal Refinement via KAOS:



- **Sub-Goal 1 (SG-1)**
  Description: Maintain: A current course calendar and materials repository
  Rationale: Lecturers and TAs need one up-to-date source for events, deadlines, and course materials so students and staff are never working from outdated informations

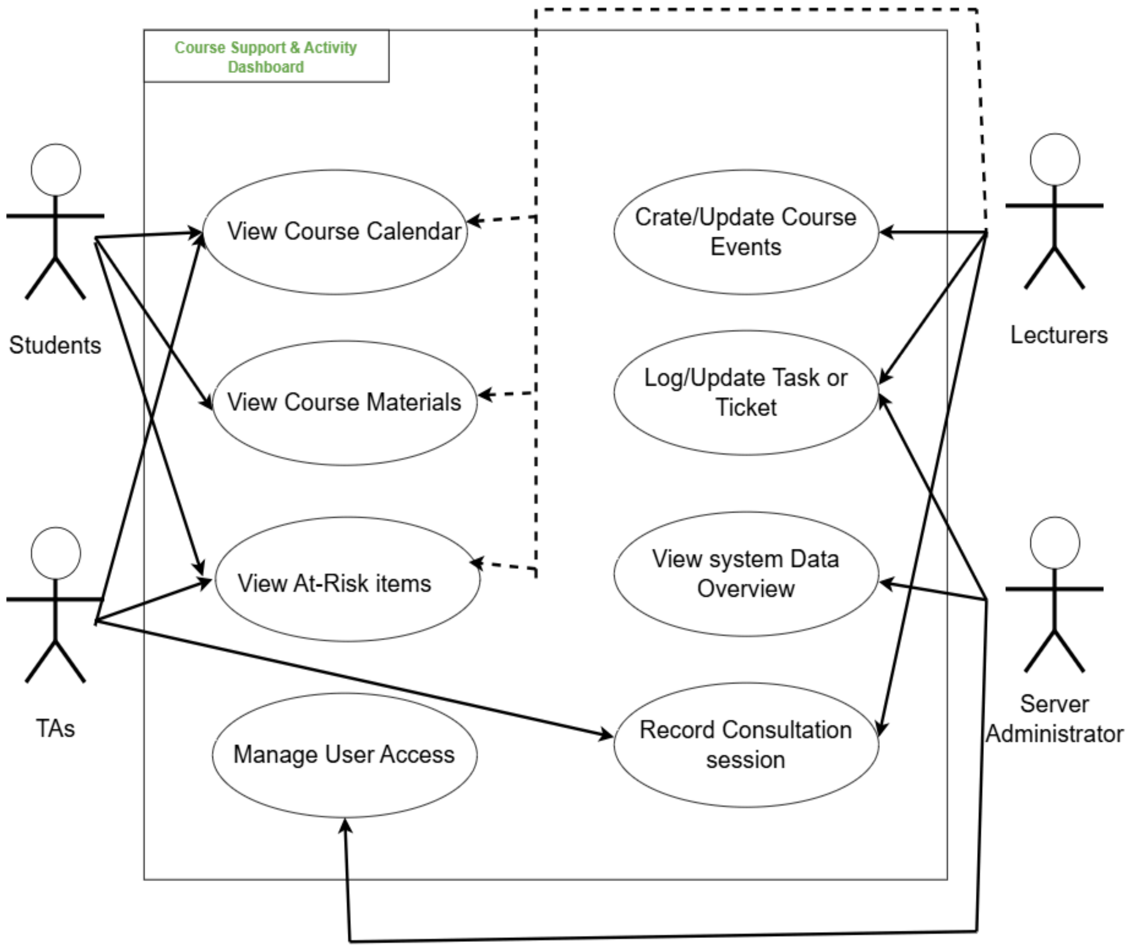
- **Sub-Goal 2 (SG-2)**
  Description: Achieve: Complete Task / Ticket details
  Rationale: Each task needs an owner, priority, and status so TAs can track and coordinate small work items without ambiguity

- **Sub-Goal 3 (SG-3)**
  Description: Achieve: Automatic flagging of at risk and overdue items
  Rationale: Lecturers need tasks nearing or past their deadline to appear automatically, rather than having to manually audit every ticket.

- **Sub-Goal 4 (SG-4)**
  Description: Maintain:A searchable consultation and support history
  Rationale: TAs need session records and common FAQs retained and searchable over the semester so recurring issues can be resolved faster.

- **Sub-Goal 5 (SG-5)**
  Description: Maintain: An accurate system data overview
  Rationale: Administrators need a single, reliable summary of system activity to monitor usage and health without checking each module individually

4. Use Case Diagram



5. Use Stories

User Story ID	Description	Acceptance Criteria
US-1	As a student i want to view the course calendar and materials so that i know the upcoming deadlines and always have an updated sturdy plans	The calendar shows the event title, type, date/time. Material shows the latest published version only
US-2	As a lecturer i want to create and update course events and materials so that students and TAs are informed with up-to-date and accurate information	Required fields(title, date/time, type) must be provided. A valid events / materials is saved and will become visible to students and TAs after it is published
US-3	As a TA i want to log and update my task / tickets so that my work is tracked with clear owner, priority and status	An authorised TA can create / update and close a ticket once the task is done. The ticket status change will be updated and shown in the ticket list immediately after saving
US-4	As a lecturer, I want to see at-risk items that were automatically flagged so that I can catch overdue or upcoming deadline tasks without having to manually check each individual item.	An authorised lecturer can access the dashboard that show the items in the upcoming or past deadlines that are automatically flagged and not yet closed
US-5	As an administrator, I want to view the summary of task, ticket, and events so that i can overview the system-wide activity and manage user access	The overview aggregates data across all modules and reflects the current state after any update. Only authorized administrators can access this view.
US-6	As a TA or Lecturer, I want to log in using my KU Google account so that only authorised department staff (@ku.th) can access the system.	Clicking “Sign in” redirects to Google OAuth2. An account outside the @ku.th domain is denied access with an error message.
US-7	As a TA or Lecturer, I want to search across all tasks/tickets, calendar events, and materials so that I can quickly locate any activity regardless of its type.	A single search returns matching tasks/tickets and materials together, showing each result's type and owner. A search with no matches shows a clear “no results” message.
US-8	As a TA or Administrator, I want to create and edit course activities and milestones (not only tickets) so that I can support the lecturer in keeping the shared calendar up to date.	An authorised TA or Administrator can create, edit, or publish a course event/material using the same required fields and validation as a Lecturer.

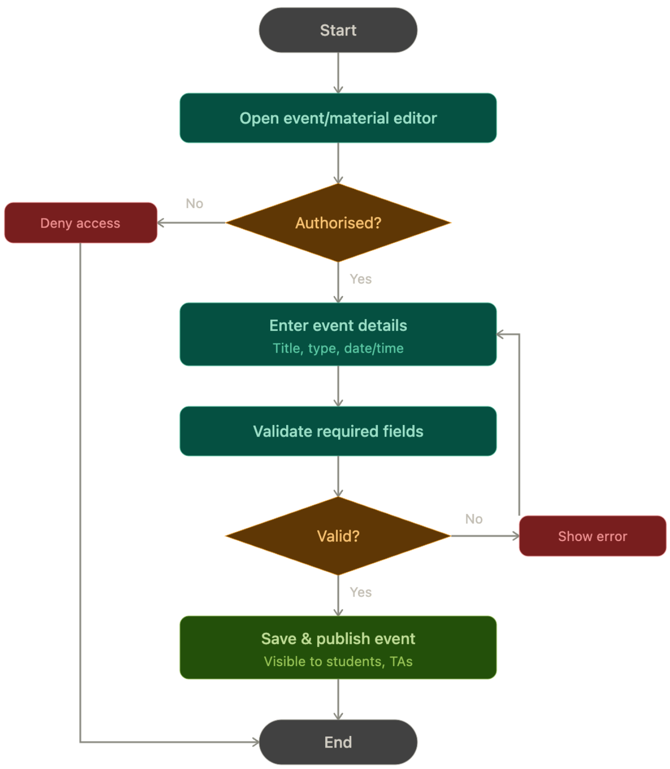
6. User Requirement Specification

ID	User Requirement
URS-1	Students and TAs shall be able to view a unified calendar of course events, lab sessions, deadlines, consultation slots, and project milestones
URS-2	Students and TAs shall be able to view current course materials and the course events, including the most recently published version.
URS-3	The system shall allow authorised TAs to create, update, and close internal tasks/tickets, including owner, priority, and status
URS-4	The system shall allow lecturers to view items automatically flagged as at-risk, such as tasks nearing or past their deadline.
URS-5	The system shall allow authorised TAs to record and search consultation sessions and support interactions across the semester.
URS-6	The system shall allow an authorised system administrator to view a consolidated summary of tasks, tickets, and events, and to manage user access.
URS-7	The system shall authenticate all users via Google OAuth2 and restrict access to accounts belonging to the @ku.th domain.
URS-8	Authorised users shall be able to search across all activities, tasks/tickets, calendar events, and materials by keyword, owner, type, status, or date.
URS-9	Authorised TAs and Administrators, in addition to Lecturers, shall be able to create and edit course activities and milestones.

7. Activity Diagram

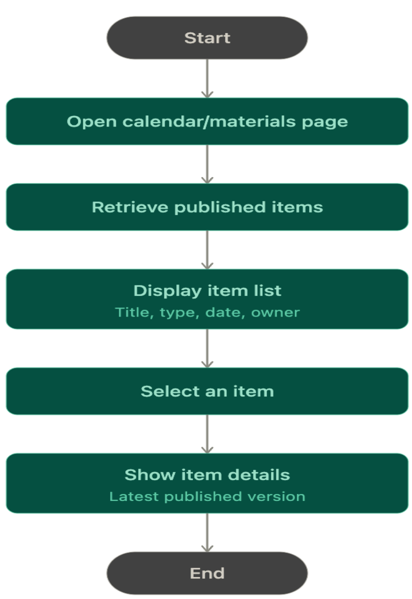
AD-1: Lecturer creates or updates a course event or material

This diagram traces the authorization check, field validation, and publishing flow used to derive SRS-3, SRS-4, and SRS-5.



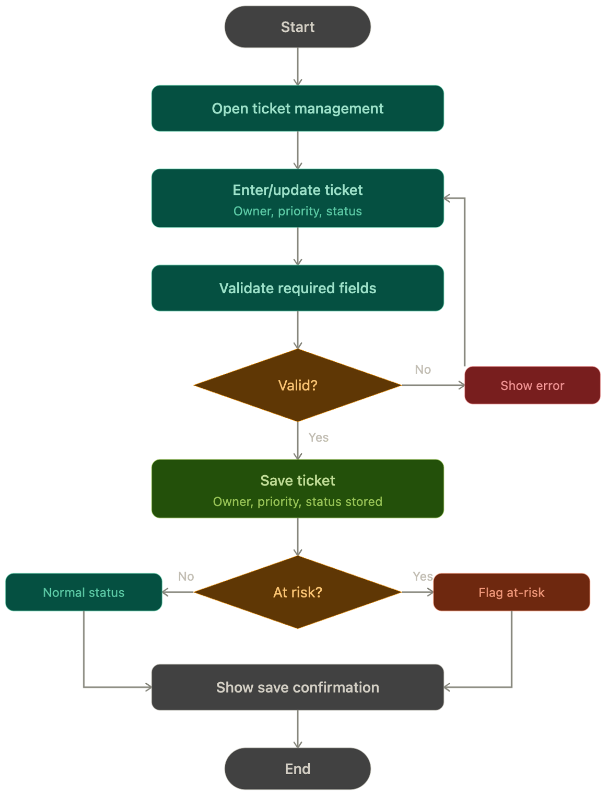
AD-2: Student/TA views course calendar and materials

This diagram traces the read-only browsing flow used to derive SRS-1 and SRS-2.



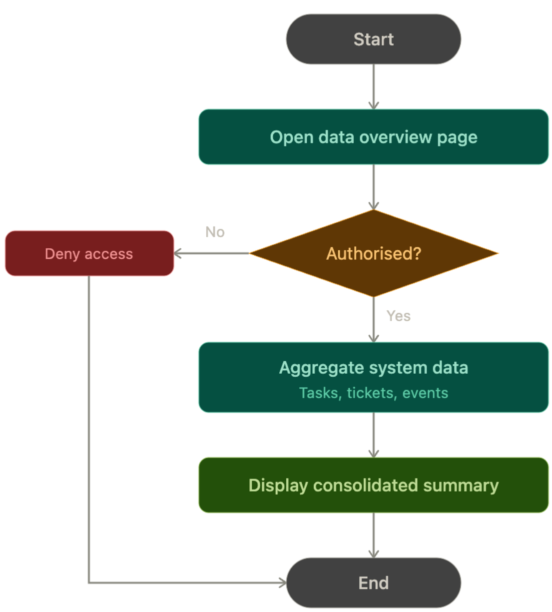
AD-3: TA logs or updates a task/ticket with automatic at-risk flagging

This diagram captures the validation and deadline-check flow used to derive SRS-5, SRS-6, and SRS-7.



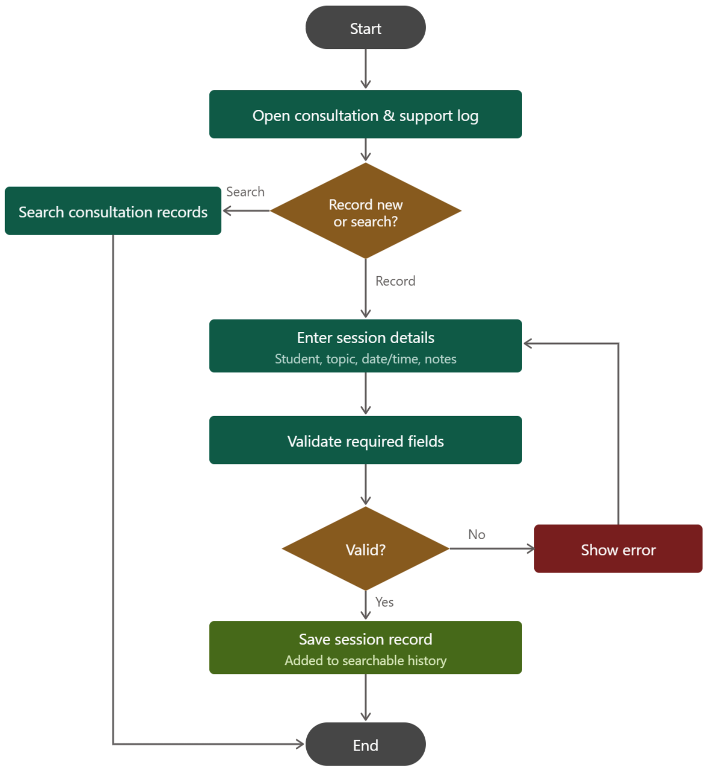
AD-4: Administrator views the consolidated system data overview

This diagram captures the authorization check and data aggregation flow used to derive SRS-8.



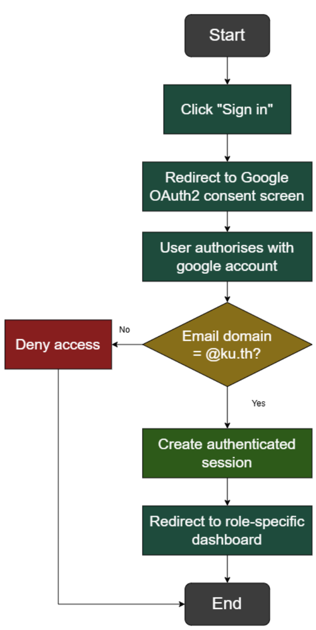
AD-5: TA records or searches a consultation/support session

This diagram traces the branching flow between recording a new consultation session and searching existing records, including field validation and error handling on the record path, used to derive SRS-9 and SRS-10.



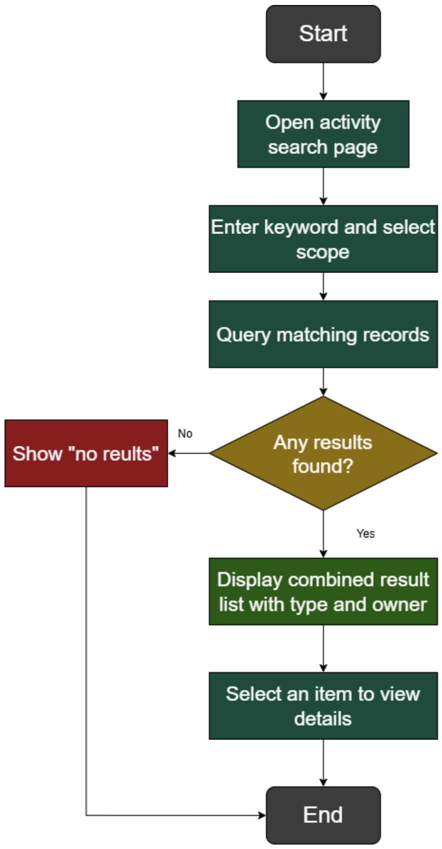
AD-6: User authenticates via Google OAuth2

This diagram traces the login, domain-check, and session flow used to derive URS-7, SRS-11, and SRS-12.



AD-7: Authorised user searches all activities

This diagram traces the combined query flow across tasks/tickets, calendar events, and materials used to derive URS-8, SRS-13, and SRS-14.



8. System Requirement Specification

ID	System Requirement
SRS-1	The system shall display only published, up-to-date version materials and events to students and TAs.
SRS-2	The system shall display the title, type, date/time, and owner for each calendar event.
SRS-3	The system shall allow only authorised Lecturers, TAs, and Administrators to create, edit, or publish course events and materials.
SRS-4	The system shall require an event title, type, and date/time before saving a new or edited calendar event.
SRS-5	The system shall display a validation message and shall not save the event or ticket when required data is missing or invalid.
SRS-6	The system shall store each ticket with an owner, priority, and status of open, in-progress, or closed.
SRS-7	The system shall automatically flag a ticket or event as at-risk when it is nearing or past its deadline and not yet closed.
SRS-8	The system shall restrict access to the consolidated system data overview to authorised system administrators only.
SRS-9	The system shall allow authorised TAs to record a consultation or support session with fields for students, topic, date/time, and notes.
SRS-10	The system shall allow authorised TAs to search consultation record by keywords, students, or date
SRS-11	The system shall redirect an unauthenticated user to Google OAuth2 sign-in before granting access to any role-specific view.
SRS-12	The system shall verify the authenticated Google account's email domain and deny access with an error message if the domain is not @ku.th.
SRS-13	The system shall allow authorised users to search tasks/tickets by keyword, owner, priority, status, or date.
SRS-14	The system shall allow authorised users to search calendar events and materials by keyword, type, or date.

9. Software Architecture

Based on the SRS for the Course Support & Activity Dashboard (Iteration 1 scope: calendar/materials management, task/ticket tracking with at-risk flagging, consultation logging, and admin data overview), the recommended architecture is a Modular Monolithic Architecture with the Model-View-Controller (MVC) pattern. This architecture creates a single deployable application unit with clearly defined, loosely linked components.

Rationale

A small team created the system to support university course operations. Adopting a distributed Microservices architecture would introduce unnecessary operational complexity, API gateways, and distributed database overhead that are unwarranted for the current iteration. Conversely, a flat monolithic design risks tight coupling as business rules, such as automatic at-risk deadline calculations (SRS-7), role-based access validation (SRS-3, SRS-8), and aggregate reporting (SRS-8).

The Modular MVC architecture balances maintainability with deployment simplicity:

- View (Presentation): Renders role-specific web interfaces for Students, TAs, Lecturers, and Administrators.
- Controller (Application & Request Handling): Validates input, enforces authentication/authorization, and routes requests to appropriate domain logic.
- Model (Business Logic & Entities): Encapsulates core business rules (like ticket state transitions, deadline checking, aggregation) and interfaces with persistence.

Component	**Layer / Type**	**Description**
Web Client Dashboard	Presentation (View)	Renders responsive interfaces for viewing calendars, managing tickets, logging consultation sessions, and presenting administrative overviews. Displays user-facing validation errors and save confirmations.

Course & Event Controller	Controller	Receives requests to publish course events and materials, verifies lecturer authorization (SRS-3), and passes validated data to the Course Model.
Task & Consultation Controller	Controller	Handles TA requests for ticket logging, status updates, and consultation session records. Manages keyword search requests (SRS-10).
Admin Overview Controller	Controller	Restricts access exclusively to authorized System Administrators (SRS-8) and coordinates data aggregation requests.
Course & Activity Domain Models	Model / Business Logic	Encapsulates entity constraints and validation rules (SRS-4, SRS-5); executes automated background logic to calculate deadlines and flag items as at-risk (SRS-7).
Authentication & Access Module	Cross-Cutting Security	Verifies user credentials and enforces role-based permissions across Lecturers, TAs, Administrators, and Students prior to write/administrative operations.
Database Gateway / ORM	Data Access Layer	Provides object-relational mapping to query, persist, and aggregate records within the underlying relational store.

10. Data Storage Technology

The recommended data storage technology is a Relational Database Management System (RDBMS), such as PostgreSQL or MySQL, acting as the centralized data store for Iteration 1.

Rationale

All domain entities defined across SRS-1 through SRS-10 : Course Events, Materials metadata, Tasks/Tickets, Consultation Logs, and User Accounts : consist of strictly structured, table data with fixed attributes.

An RDBMS is selected because:

- Relational Integrity & Schema Enforcement: Entities maintain clear relational foreign keys (linking a ticket or consultation log to a specific TA/student user record, or course materials to specific course modules). Schema constraints ensure mandatory fields (title, event type, date/time, ticket priority) cannot be omitted (SRS-4, SRS-5).
- ACID Transactional Consistency: Updating ticket statuses, logging support notes, and evaluating at-risk flags require transactional consistency so concurrent updates by multiple TAs and Lecturers are immediately reflected on the shared dashboard without race conditions (SRS-6, SRS-7).
- Relational Querying & Aggregation: The Administrator Data Overview (SRS-8) and TA search queries (SRS-10) rely heavily on SQL filtering, indexing, and aggregate functions across tasks, events, and consultation logs.
- No Requirement for Unstructured Storage: Iteration 1 does not involve dynamic unstructured JSON documents or high-volume real-time streams that would require a NoSQL store.

Component	**Data Storage Technology**	**Description**
Course Support Database	RDBMS (PostgreSQL / MySQL)	Normalized relational tables storing Users, CourseEvents, CourseMaterials, Tickets, and ConsultationLogs. Enforces referential integrity, indexes timestamps/keywords for search (SRS-10), and persists status flags (SRS-6, SRS-7).

11. Development Environment

The recommended development and deployment environment is a Virtual Machine (VM) / containerised environment (e.g., using Docker), rather than bare metal.

Rationale

The Course Support & Activity Dashboard is developed by a multi-member team (Guild Board) and must be accessible via standard web browsers by various target user roles (Lecturers, TAs, Students, and System Administrators) across different devices. A containerised/VM-based environment packages the web application, backend services, dependencies, and the relational database into portable, reproducible containers.

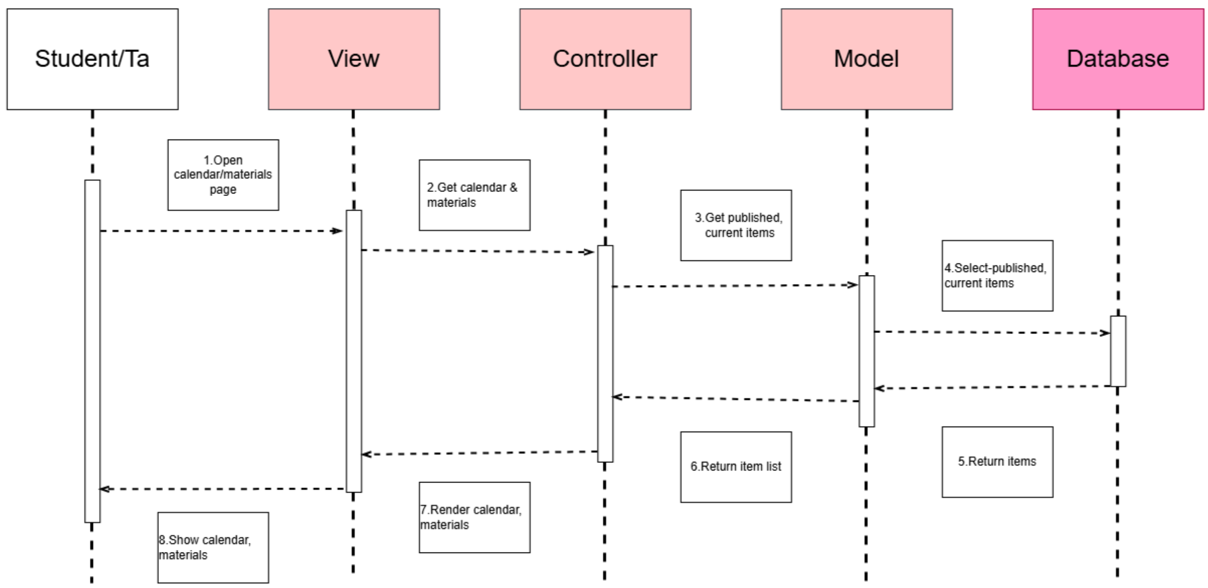
This setup allows team members to develop and test features locally across different operating systems without encountering "works on my machine" dependency conflicts. Deploying via containers also decouples the application and database from the underlying physical server or departmental host machine, simplifying updates, environment rollbacks, and future migrations without requiring complex bare-metal configuration.

Component	Development Environment	Description
Application Server (Backend / Web App)	Docker container	Runs the core application logic, API endpoints, authentication, and view routing (e.g., calendar management, ticket tracking, and admin aggregation) in an isolated, reproducible container image.
Course & Support Database (RDBMS)	Docker container (via Docker Compose)	Runs the relational database storing course events, materials metadata, tasks/tickets, consultation logs, and user access records, orchestrated alongside the application container using Docker Compose.
Local Development Machine	Local Docker Desktop / Virtual Machine	Used by team members to develop, test, and verify features (e.g., at-risk flagging logic, search queries) before staging or production deployment, ensuring complete parity with the host environment.

12. Sequence Diagrams for All SRS

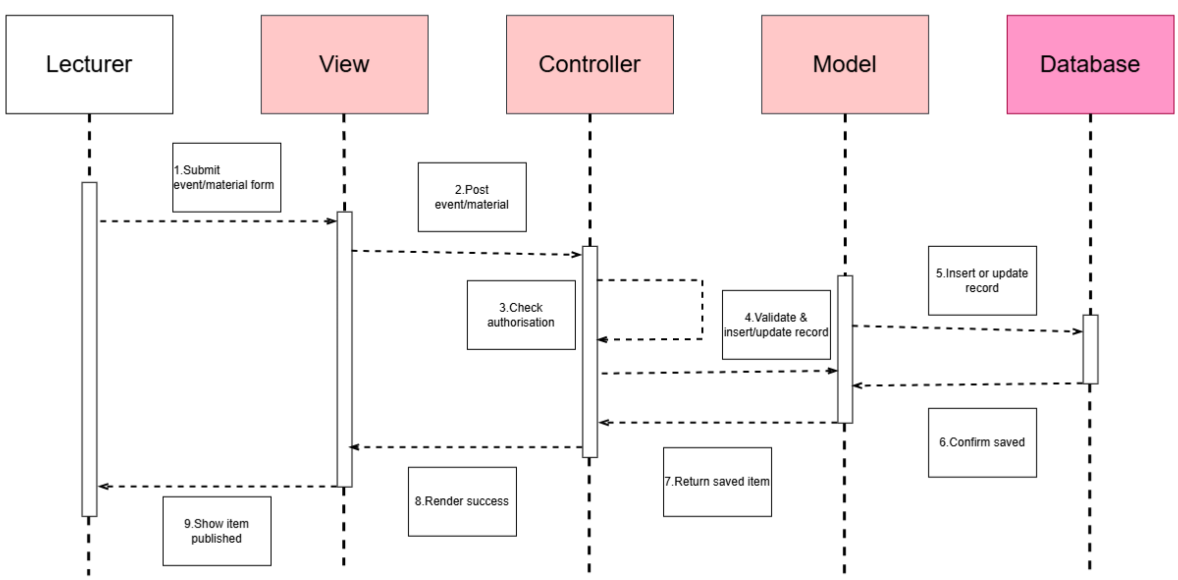
SQD-1: Student/TA Views Course Calendar & Materials

Traced requirements: SRS-1, SRS-2



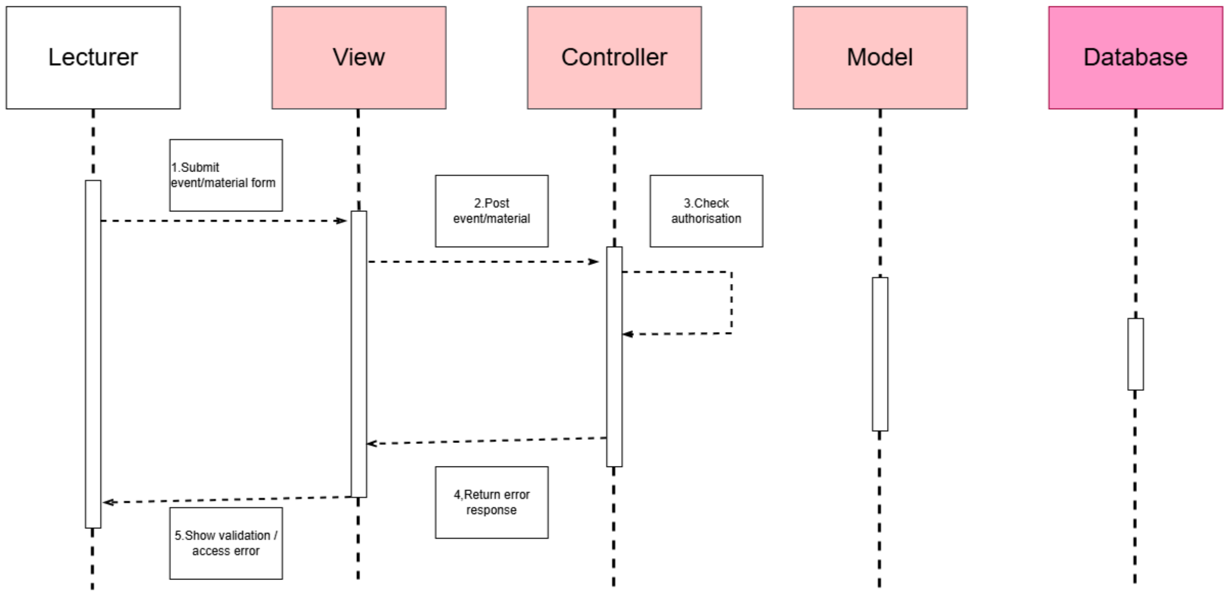
SQD-2: Lecturer Creates/Publishes Course Event or Material (Success Path)

Traced requirements: SRS-3, SRS-4



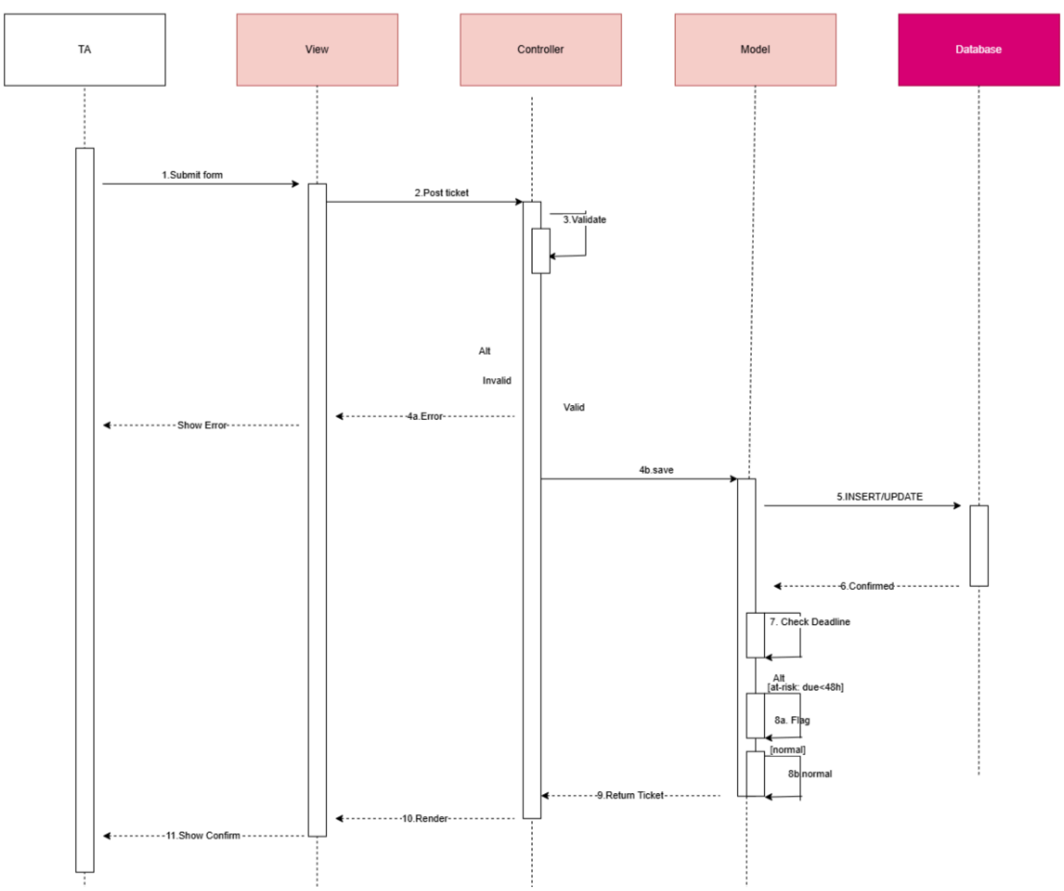
SQD-3: Lecturer Creates/Publishes Course Event or Material (Error Path)

Traced requirements: SRS-3, SRS-5



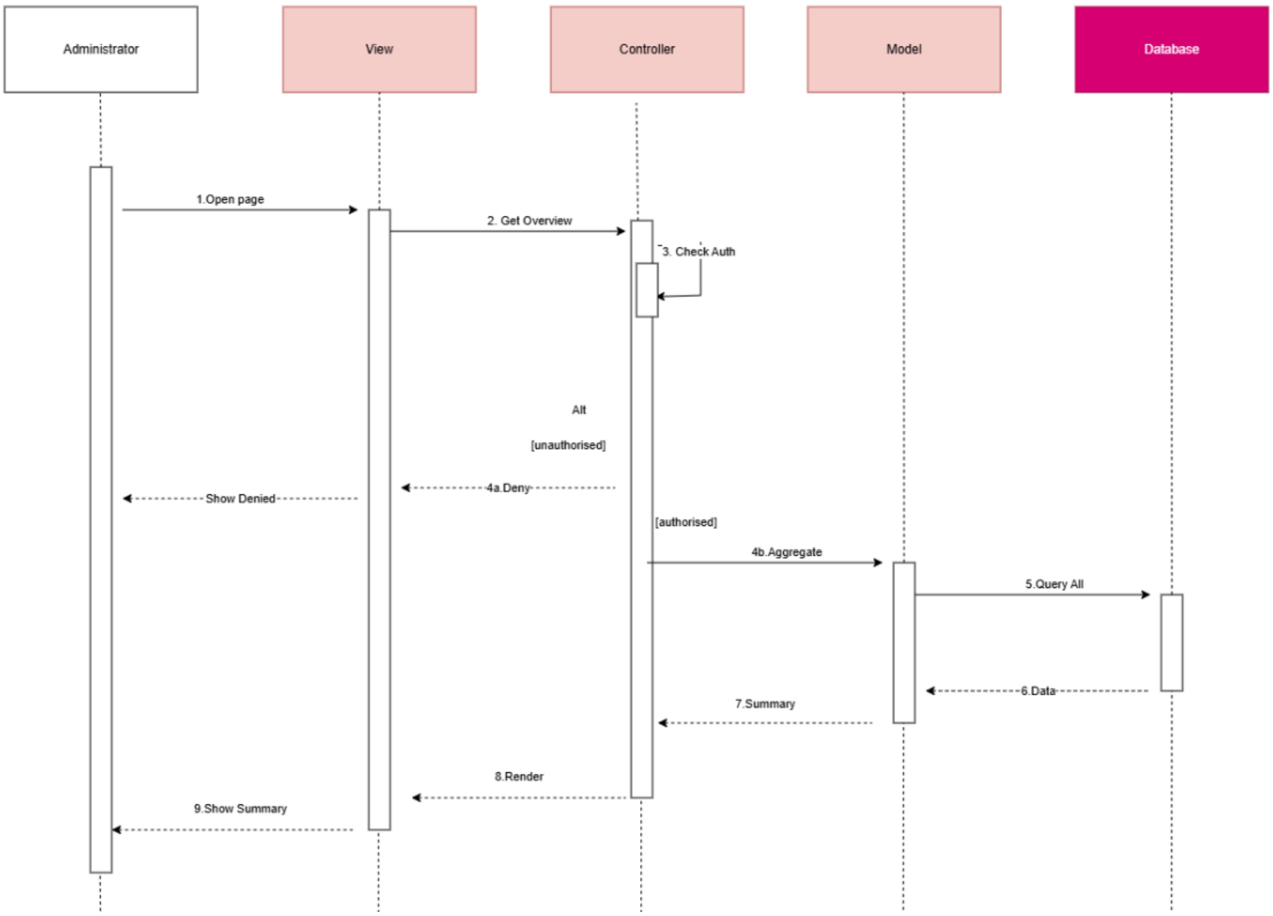
SQD-4: TA Logs/Updates a Task or Ticket with At-Risk Flagging

Traced requirements: SRS-5, SRS-6, SRS-7



SQD-5: Administrator Views Consolidated Data Overview

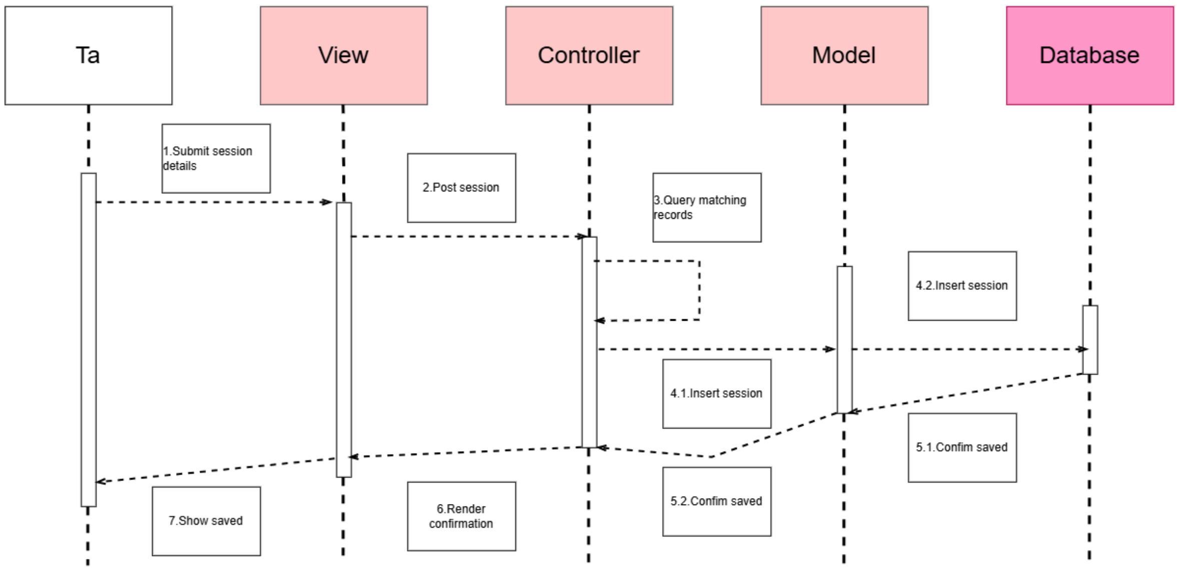
Traced requirements: SRS-8



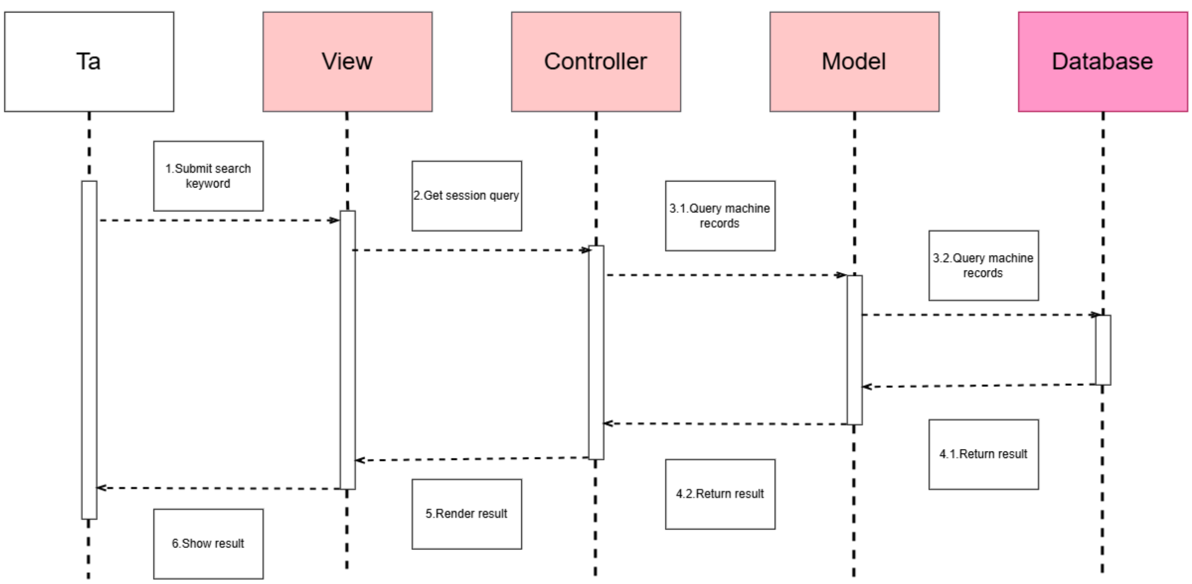
SQD-6: TA Records/Searches a Consultation Session

Traced requirements: SRS-9, SRS-10

6a: Record Branch:

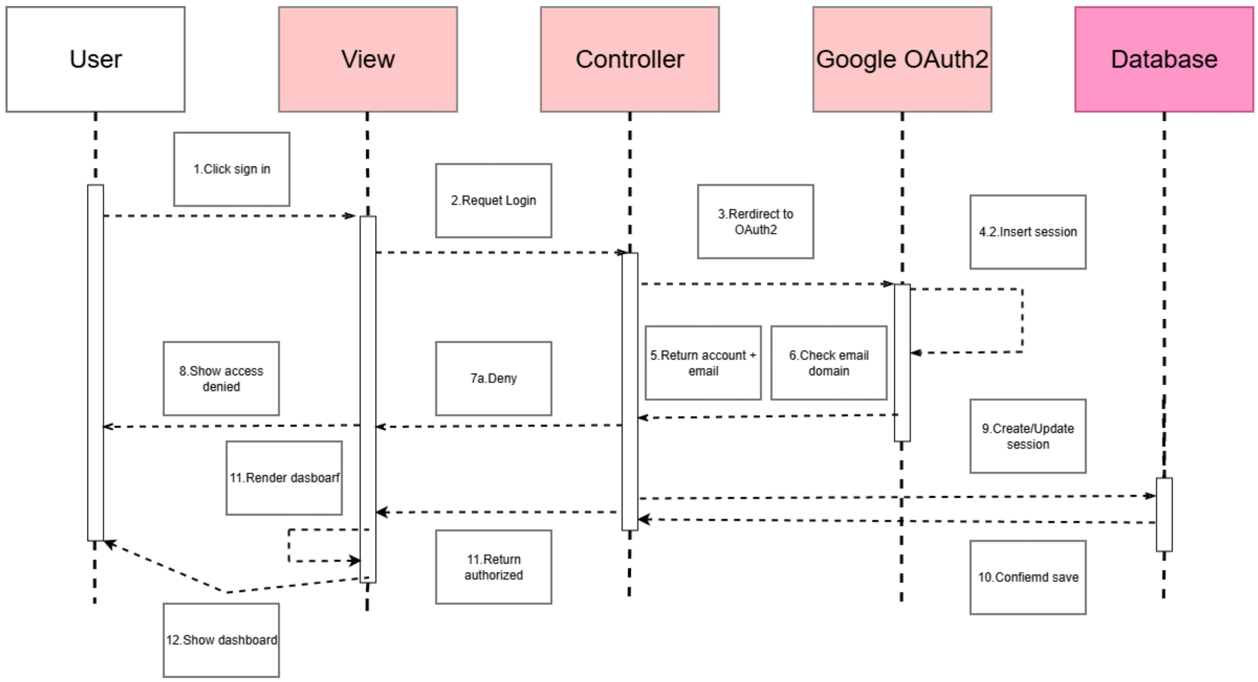


6b: Search Branch:



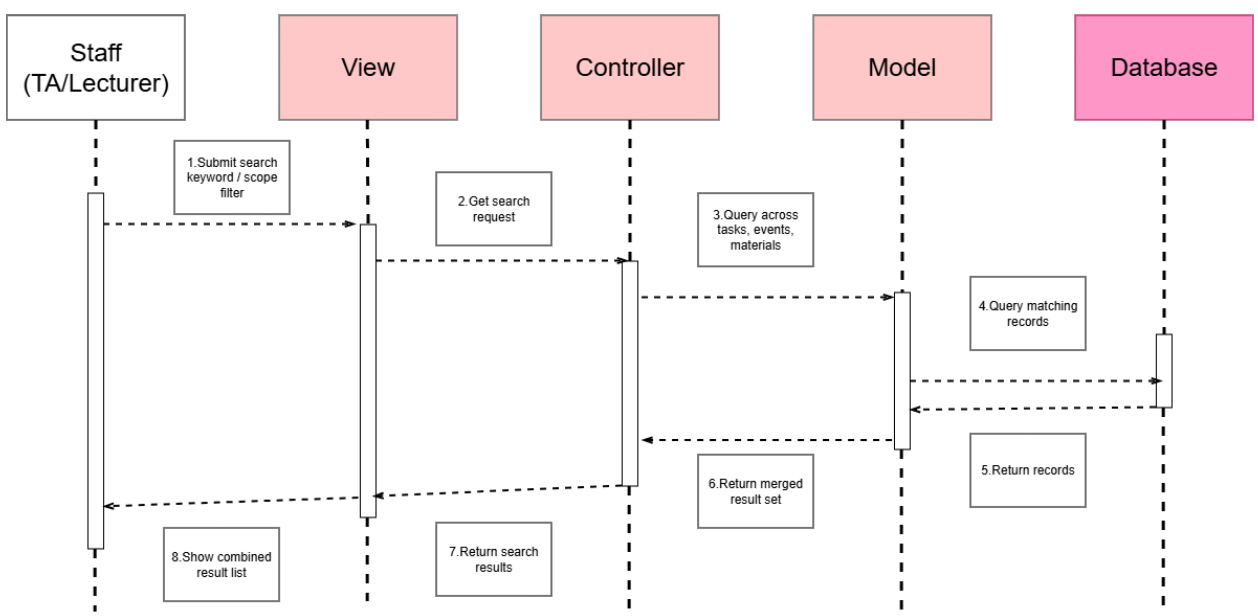
SQD-7: User Authenticates via Google OAuth2

Traced requirements: SRS-11, SRS-1



SQD-8: Authorised Staff Searches All Activities

Traced requirements: SRS-13, SRS-14T



13. Traceability Matrix

Sub-Goal	Use Case	User Story	URS	Activity Diagram	SRS
SG-1	UC-1 View Course Calendar and Materials	US-1	URS-1, URS-2	AD-2	SRS-1, SRS-2
SG-2	UC-2 Log/Update Task or Ticket	US-3	URS-3	AD-3	SRS-3, SRS-4, SRS-5, SRS-6
SG-3	UC-3 View At-Risk Items	US-4	URS-4	AD-3	SRS-7
SG-4	UC-4 Record Consultation Section	US-3	URS-5	AD-5	SRS-9, SRS-10
SG-5	UC-5 View System Data Overview	US-5	URS-6	AD-4	SRS-8
SG-5 (extended)	UC-7 Manage User Access	US-10	URS-6	AD-4	SRS-8
SG-5 (extended)	UC-6 Authenticate via Google Auth2	US-6	URS-7	AD-6	SRS-11, SRS-12
SG-1 / SG-2 (extended)	UC-7 Search all activities	US-7	URS-8	AD-7	SRS-13, SRS-14
SG-2 (extended)	UC-8 TAs / Admins Creates or Edits Activities	US-8	URS-9	AD-1	SRS-3

From SRS to Sequence Diagram

SRS	**SQD**
SRS-1	SQD-1
SRS-2	SQD-1
SRS-3	SQD-2, SQD-3
SRS-4	SQD-2
SRS-5	SQD-3, SQD-4
SRS-6	SQD-4
SRS-7	SQD-4
SRS-8	SQD-5
SRS-9	SQD-6
SRS-10	SQD-6
SRS-11	SQD-7
SRS-12	SQD-7
SRS-13	SQD-8
SRS-14	SQD-8